

p1

Exported 2/21/2026, 10:10:55 AM

For all problems, the heat rod model is imported from homework 1 (filename heat_rod.py). I'm attaching heat_rod.py code in the Appendix, without the pages assigned for each problem, for clarity.

Similarly, mcmc.py provides the code for the standard Metropolis Hastings algorithm. am_mcmc.py includes the adaptive method for problem 2. gibbs_mcmc.py includes the gibbs sampling step for problem 3. Since these three codes are used for multiple subproblems, the codes are attachedd to the bottom of the pdf, rather than assigned to each specific problem. For all codes/results, please simply view my github repo at:

<https://github.com/rx9933/ase379l/tree/main/hw6>

1.1

In [4]:

```
import os, sys, numpy as np, matplotlib.pyplot as plt
sys.path.append('../hw1')
import heat_rod as hr
from mcmc import *
from scipy.io import loadmat

data = loadmat("HW06_Problem1.mat")
samples, q_map, posterior_values, acceptance_rate =
run_metropolis_for_heat_equation()

# Save results
np.savez('problem1.1_results.npz',
        samples=samples,
        q_map=q_map,
        posterior_values=posterior_values,
        acceptance_rate=acceptance_rate)

print(f"\nResults saved to 'problem1.1_results.npz'")
```

```

Shape of observations: (15,)
Initial test - r(q0, q0) = 1.000000

Running Metropolis algorithm for 1000 iterations...
Proposal distribution: N(q_prev, D) with D = diag([0.01, 2e-10])
Running Metropolis-Hastings MCMC...
  Completed 200/1000 iterations, acceptance rate: 0.719
  Completed 400/1000 iterations, acceptance rate: 0.759
  Completed 600/1000 iterations, acceptance rate: 0.783
  Completed 800/1000 iterations, acceptance rate: 0.796
  Completed 1000/1000 iterations, acceptance rate: 0.799

Final acceptance rate: 0.799

Computing posterior values for MAP estimate...

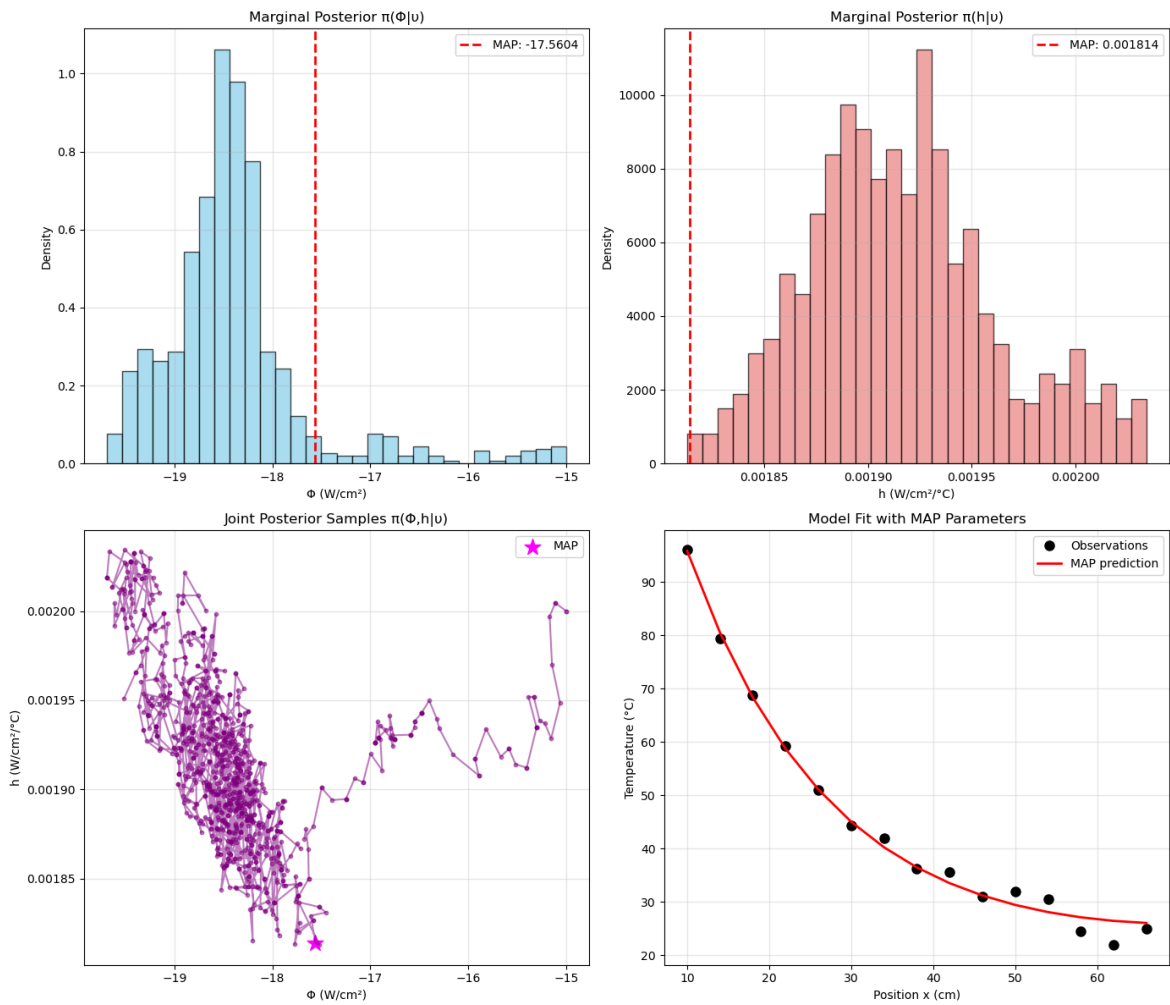
=====
PROBLEM 1.1 RESULTS
=====

MAP Estimate (q_MAP):
   $\Phi = -17.560427 \text{ W/cm}^2$ 
   $h = 0.001814 \text{ W/cm}^2/\text{°C}$ 
   $\pi(u|q_{\text{MAP}})\pi\theta(q_{\text{MAP}}) = 5.61\text{e-}03$ 

Posterior Distribution  $\pi(q|u)$  Summary:
=====
 $\Phi$ :
  Mean = -18.395988 W/cm2
  Std  = 0.747602 W/cm2
  95% CI = [-19.464980, -15.928881]

h:
  Mean = 1.916442e-03 W/cm2/°C
  Std  = 4.574893e-05 W/cm2/°C
  95% CI = [1.836325e-03, 2.018715e-03]

```



Results saved to 'problem1.1_results.npz'

In [5]:

```
fig = plt.figure(figsize=(14, 10))
ax = fig.add_subplot(111, projection='3d')

scatter = ax.scatter(samples[:, 0], samples[:, 1], posterior_values,
                      c=posterior_values, cmap='viridis',
                      s=20, alpha=0.6, marker='o')

# Mark the MAP estimate
map_idx = np.argmax(posterior_values)
ax.scatter([samples[map_idx, 0]], [samples[map_idx, 1]],
           [posterior_values[map_idx]],
           color='red', s=200, marker='*', label=f'MAP Estimate')

ax.scatter([samples[0, 0]], [samples[0, 1]], [posterior_values[0]],
           color='blue', s=200, marker='*', label=f'Init Estimate')

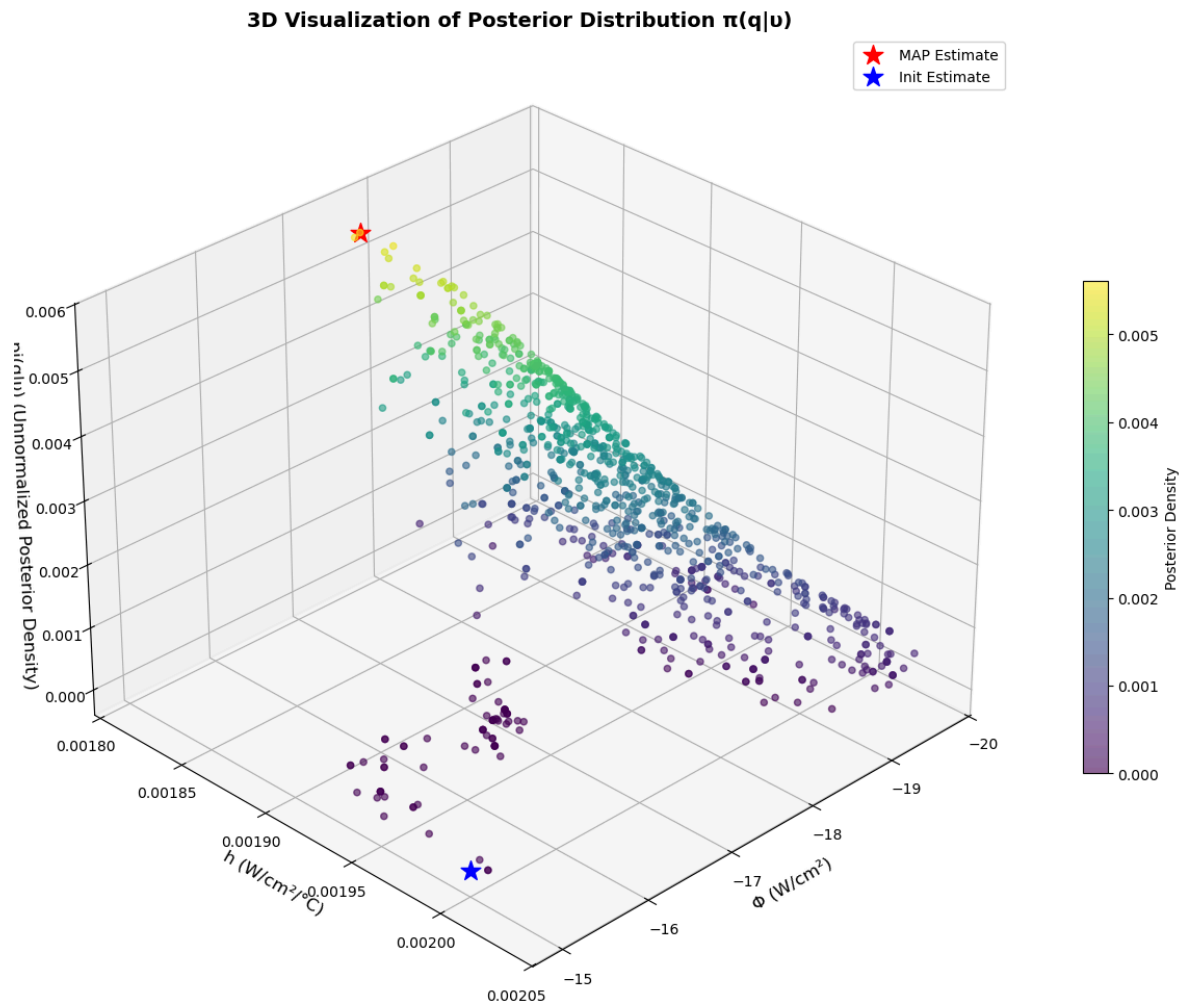
# Add labels and title
ax.set_xlabel('Φ (W/cm²)', fontsize=12)
ax.set_ylabel('h (W/cm²/°C)', fontsize=12)
ax.set_zlabel('π(q|u) (Unnormalized Posterior Density)', fontsize=12)
ax.set_title('3D Visualization of Posterior Distribution π(q|u)',
             fontsize=14, fontweight='bold')

# Add colorbar
cbar = fig.colorbar(scatter, ax=ax, shrink=0.5, aspect=20)
cbar.set_label('Posterior Density', fontsize=10)

# Add legend
ax.legend()

# Set viewing angle for better initial view
ax.view_init(elev=30, azimuth=45)

plt.tight_layout()
plt.savefig('posterior_3d_scatter.png', dpi=150)
plt.show()
```



In [6]:

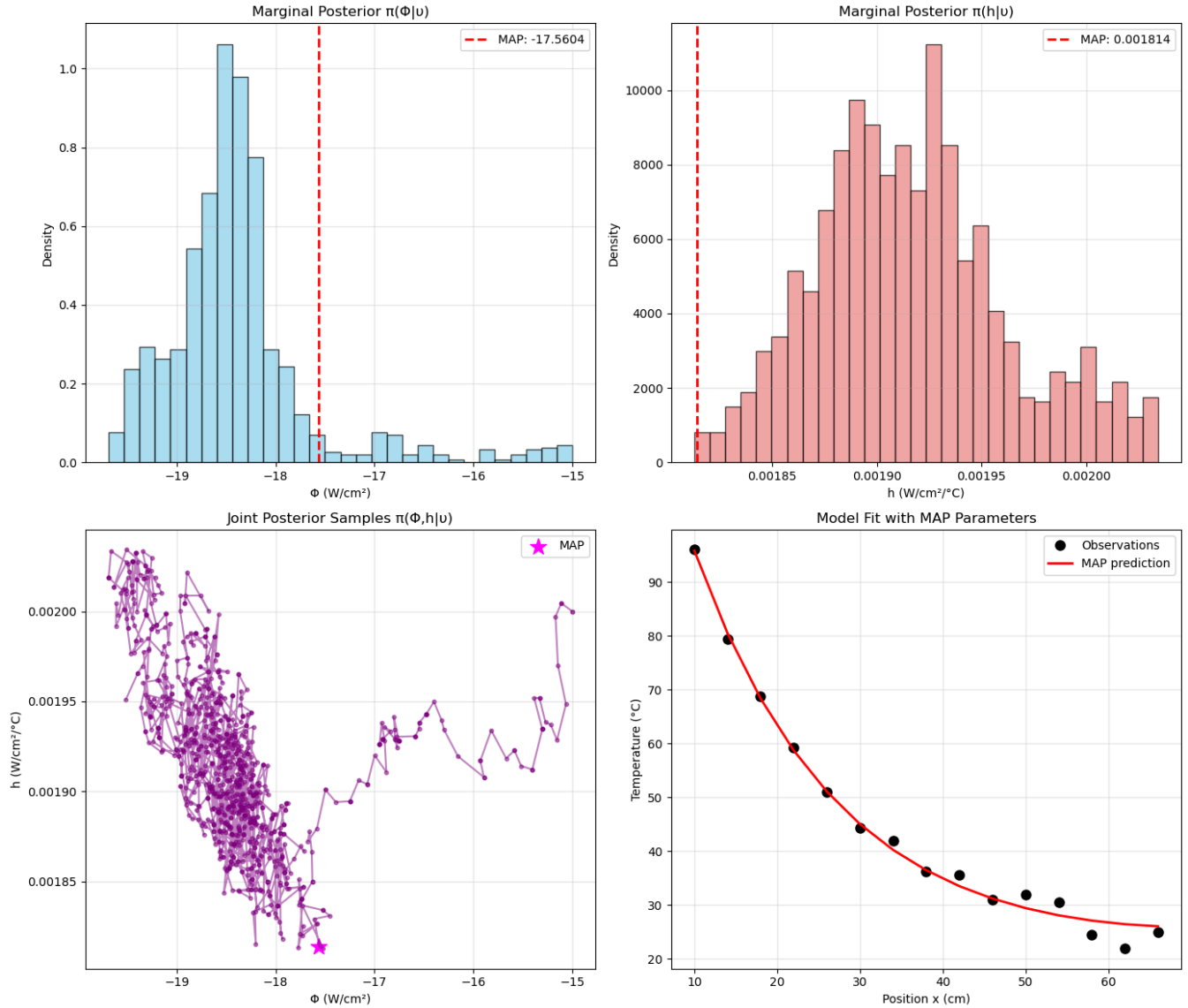
```
max(posterior_values)
```

```
np.float64(0.005609520242829489)
```

$${}^{22}q_{MAP} = [\Phi = -17.56 W / cm^2, h = 0.0018 W / cm^2 / ^\circ C]$$

$$\pi(q_{MAP} | \nu) \approx 0.0056$$

1.2



After burn-in, the chain gradually reaches a stationary distribution, but that does not mean the samples are independent (i.i.d). An MCMC chain satisfies the Markov property,

$$q^{(k+1)} \sim P(\cdot | q^{(k)}),$$

so each sample depends directly on the previous one. Even after convergence to the target distribution,

$$q^{(k)} \sim \pi(q),$$

the samples are identically distributed (after burn-in) but not independent. In particular, there is generally nonzero serial correlation,

$$\text{Cov}(q^{(k)}, q^{(k+1)}) \neq 0.$$

Thus, after sufficient burn-in we have draws from the correct stationary distribution $\pi(q)$, but that does not mean $q^{(k)}$ and $q^{(k+1)}$ are independent, identically distributed.

The chain retains autocorrelation because of its sequential construction. This matters because correlated samples reduce the effective sample size (ESS), so $ESS < N$, and variance estimates must account for autocorrelation. Thinning is can be used to reduce correlation (e.g., save only the r^{th} sample to minimize correlation between samples). Thus, burn-in guarantees convergence to the stationary distribution, but it does not imply independence or i.i.d. samples; MCMC produces correlated draws from the correct (of course approximated) target distribution (eventually).

1.3

Acceptance ratio = 0.799 (near/at 1000/1000 sample)

1.4

Large D

In [7]:

```
d2 = np.array([[1e-2, 0],
               [0, 2e-10]]) * 1000
samples, q_map, posterior_values, acceptance_rate =
run_metropolis_for_heat_equation(D = d2)

# Save results
np.savez('problem1.4_large_D.npz',
        samples=samples,
        q_map=q_map,
        posterior_values=posterior_values,
        acceptance_rate=acceptance_rate)
```

Shape of observations: (15,)

Initial test - $r(q_0, q_0) = 1.000000$

Running Metropolis algorithm for 1000 iterations...

Proposal distribution: $N(q_{\text{prev}}, D)$ with $D = \text{diag}([10.0, 2.0000000000000002e-07])$

Running Metropolis-Hastings MCMC...

Completed 200/1000 iterations, acceptance rate: 0.060

Completed 400/1000 iterations, acceptance rate: 0.063

Completed 600/1000 iterations, acceptance rate: 0.052

Completed 800/1000 iterations, acceptance rate: 0.054

Completed 1000/1000 iterations, acceptance rate: 0.049

Final acceptance rate: 0.049

Computing posterior values for MAP estimate...

PROBLEM 1.1 RESULTS

MAP Estimate (q_{MAP}):

$\Phi = -17.719167 \text{ W/cm}^2$

$h = 0.001837 \text{ W/cm}^2/\text{°C}$

$\pi(u|q_{\text{MAP}})\pi_0(q_{\text{MAP}}) = 5.03e-03$

Posterior Distribution $\pi(q|u)$ Summary:

Φ :

Mean = $-18.092433 \text{ W/cm}^2$

Std = 1.612962 W/cm^2

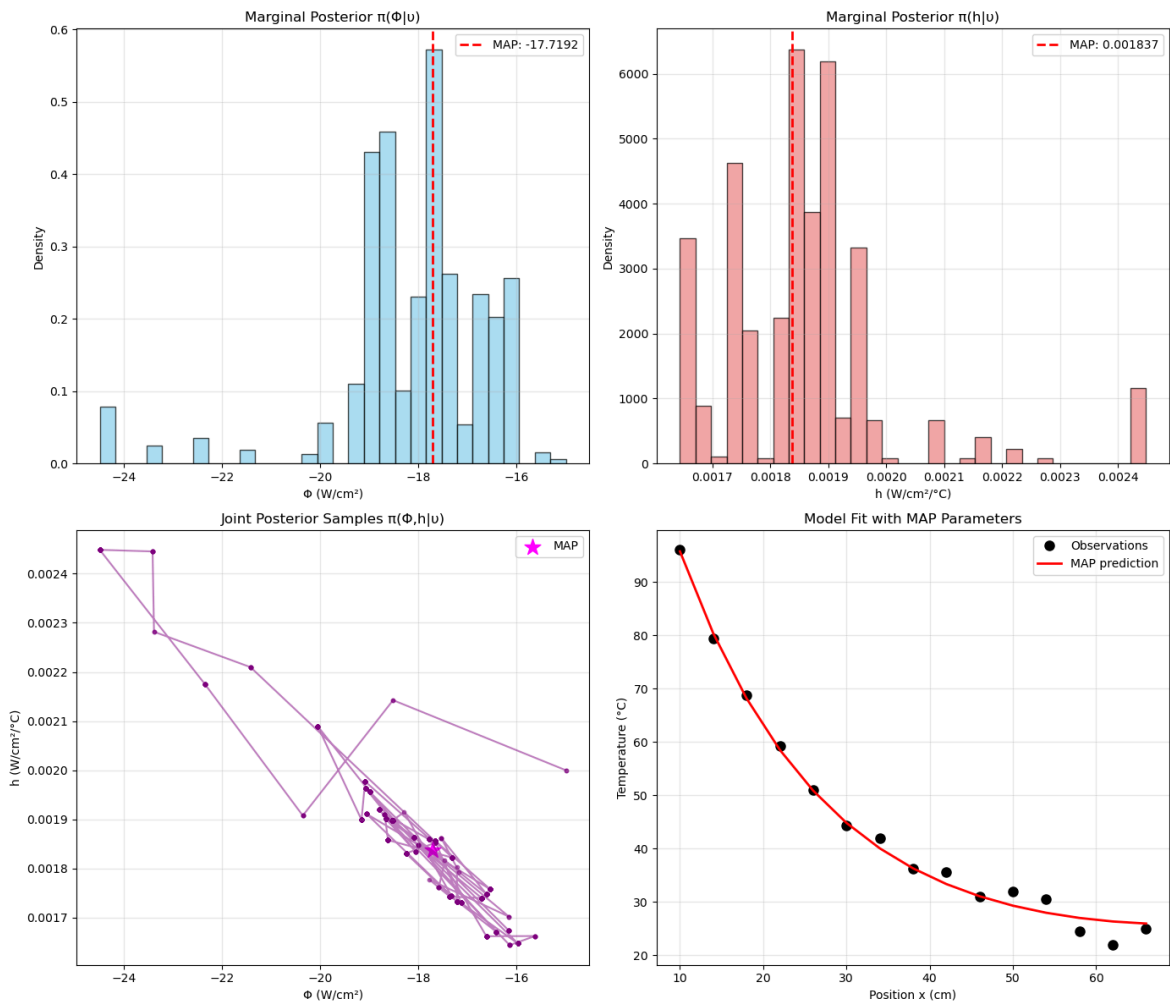
95% CI = $[-23.436418, -15.978914]$

h :

Mean = $1.860009e-03 \text{ W/cm}^2/\text{°C}$

Std = $1.514691e-04 \text{ W/cm}^2/\text{°C}$

95% CI = $[1.647691e-03, 2.445395e-03]$



Small D

In [8]:

```
d3 = np.array([[1e-2, 0],
               [0, 2e-10]]) / 1000
samples, q_map, posterior_values, acceptance_rate =
run_metropolis_for_heat_equation(D = d3)

# Save results
np.savez('problem1.4_small_D.npz',
        samples=samples,
        q_map=q_map,
        posterior_values=posterior_values,
        acceptance_rate=acceptance_rate)
```

```

Shape of observations: (15,)
Initial test - r(q0, q0) = 1.000000

Running Metropolis algorithm for 1000 iterations...
Proposal distribution: N(q_prev, D) with D = diag([1e-05, 2e-13])
Running Metropolis-Hastings MCMC...
    Completed 200/1000 iterations, acceptance rate: 0.905
    Completed 400/1000 iterations, acceptance rate: 0.927
    Completed 600/1000 iterations, acceptance rate: 0.927
    Completed 800/1000 iterations, acceptance rate: 0.935
    Completed 1000/1000 iterations, acceptance rate: 0.932

Final acceptance rate: 0.932

Computing posterior values for MAP estimate...

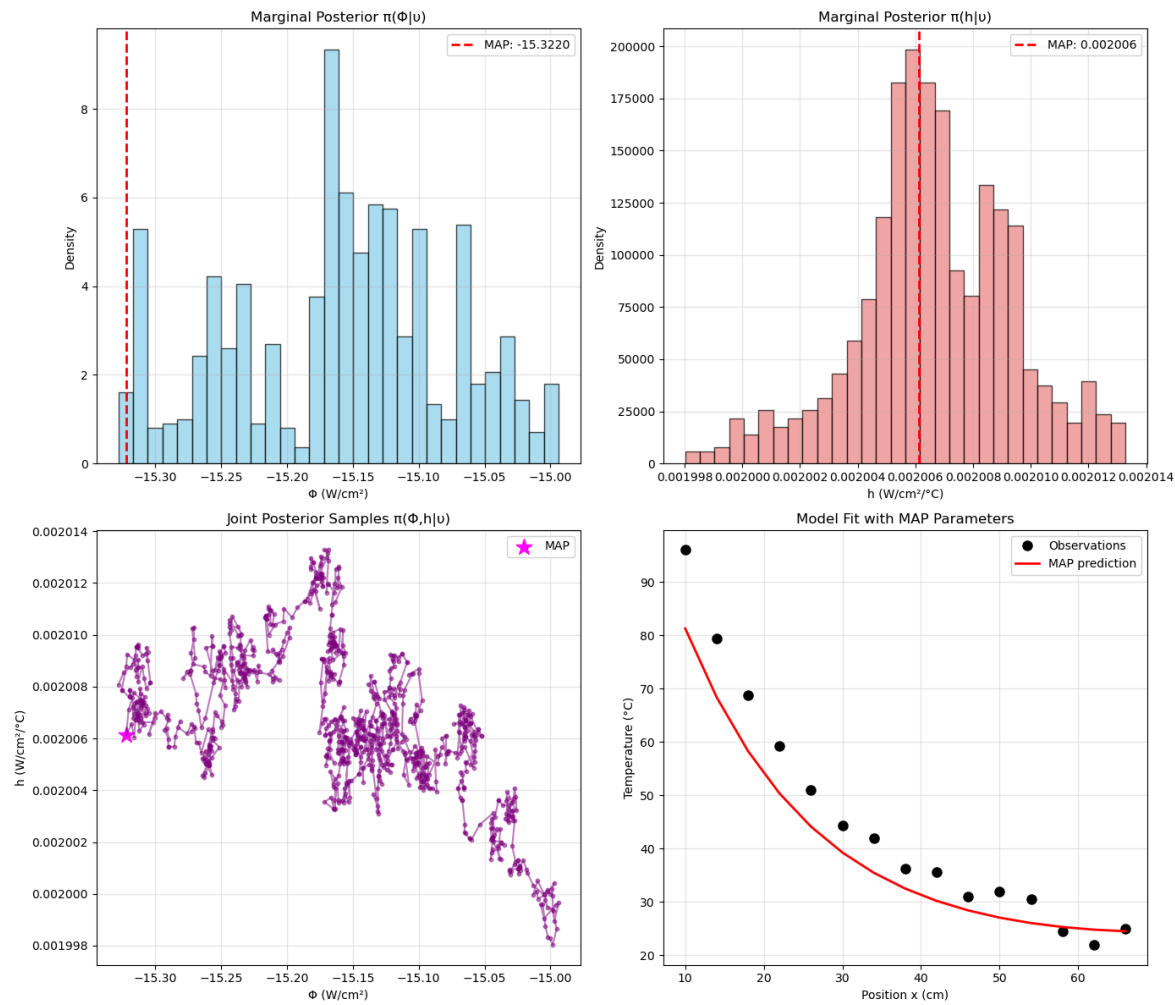
=====
PROBLEM 1.1 RESULTS
=====

MAP Estimate (q_MAP):
     $\Phi = -15.322028 \text{ W/cm}^2$ 
     $h = 0.002006 \text{ W/cm}^2/\text{°C}$ 
     $\pi(u|q_{\text{MAP}})\pi_0(q_{\text{MAP}}) = 6.64\text{e-}41$ 

Posterior Distribution  $\pi(q|u)$  Summary:
=====
 $\Phi$ :
    Mean = -15.159361 W/cm2
    Std  = 0.082691 W/cm2
    95% CI = [-15.315041, -15.010574]

h:
    Mean = 2.006650e-03 W/cm2/°C
    Std  = 2.725994e-06 W/cm2/°C
    95% CI = [2.000325e-03, 2.012206e-03]

```



Off-Diagonal

In [9]:

```
d4 = np.array([[1e-2, 1e-5],
               [0, 2e-10]])
samples, q_map, posterior_values, acceptance_rate =
run_metropolis_for_heat_equation(D = d4)

# Save results
np.savez('problem1.4_off_D.npz',
         samples=samples,
         q_map=q_map,
         posterior_values=posterior_values,
         acceptance_rate=acceptance_rate)
```

```
Shape of observations: (15,)
Initial test - r(q0, q0) = 1.000000

Running Metropolis algorithm for 1000 iterations...
Proposal distribution: N(q_prev, D) with D = diag([0.01, 2e-10])
Running Metropolis-Hastings MCMC...
  Completed 200/1000 iterations, acceptance rate: 0.362
  Completed 400/1000 iterations, acceptance rate: 0.311
  Completed 600/1000 iterations, acceptance rate: 0.294
```

```
/home/mili/ase3791/hw6/mcmc.py:154: RuntimeWarning: covariance is not
symmetric positive-semidefinite.
  return np.random.multivariate_normal(q_current, D)
```

Completed 800/1000 iterations, acceptance rate: 0.293
 Completed 1000/1000 iterations, acceptance rate: 0.284

Final acceptance rate: 0.284

Computing posterior values for MAP estimate...

=====

PROBLEM 1.1 RESULTS

=====

MAP Estimate (q_{MAP}):

$\Phi = -15.657959 \text{ W/cm}^2$

$h = 0.001632 \text{ W/cm}^2/\text{°C}$

$\pi(u|q_{\text{MAP}})\pi\theta(q_{\text{MAP}}) = 1.30\text{e-}03$

Posterior Distribution $\pi(q|u)$ Summary:

=====

Φ :

Mean = $-15.533098 \text{ W/cm}^2$

Std = 0.077003 W/cm^2

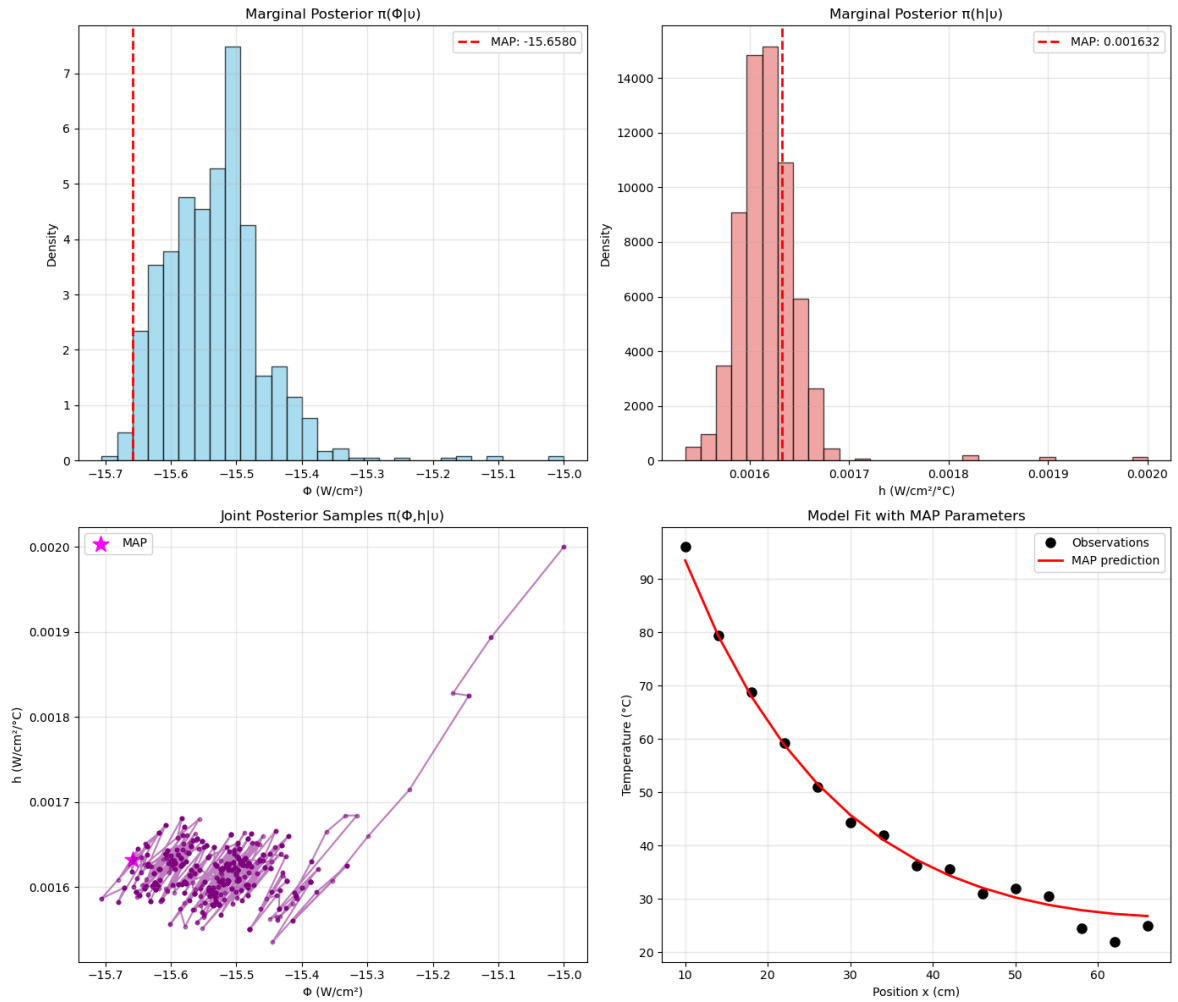
95% CI = $[-15.651538, -15.386582]$

h :

Mean = $1.618345\text{e-}03 \text{ W/cm}^2/\text{°C}$

Std = $3.463688\text{e-}05 \text{ W/cm}^2/\text{°C}$

95% CI = $[1.567229\text{e-}03, 1.666204\text{e-}03]$



In manipulating the proposal covariance matrix D and initial guess q_0 for the Metropolis algorithm, we notice that the diagonal elements of D , which control the step sizes for Φ and h respectively, have a direct impact on the acceptance rate and mixing behavior of the chain. When D is too small, the proposal steps become very small, leading to high acceptance rates but extremely poor mixing—the chain explores the parameter space very slowly and exhibits high autocorrelation, requiring many more iterations to adequately sample the posterior distribution. Conversely, when D is too large, the proposals frequently land in regions of extremely low posterior probability, causing the acceptance rate to plummet and the chain to become "stuck" for long periods, again resulting in inefficient exploration. The optimal D balances these extremes, typically yielding acceptance rates between 20-40% and allowing the chain to explore the posterior efficiently. Regarding the initial guess q_0 , starting far from the high-probability region (e.g., $\Phi = -5$ or $h = 0.01$) simply extends the burn-in period as the chain must wander toward the posterior mode, but does not affect the final stationary distribution once convergence is achieved (as long as the extended burn-in is removed).

The use of logarithms in the likelihood and prior calculations effectively prevents numerical underflow issues that could otherwise arise when multiplying many small probability densities, particularly when evaluating proposals far from the posterior mode. This logarithmic approach ensures numerical stability across a wide range of D values and initial conditions, allowing the algorithm to handle poor proposals without encountering floating-point precision limitations.

1.5

In [10]:

```
samples, q_map, posterior_values, acceptance_rate =  
run_metropolis_for_heat_equation(M = 100000)  
  
# Save results  
np.savez('problem1.5.npz',  
         samples=samples,  
         q_map=q_map,  
         posterior_values=posterior_values,  
         acceptance_rate=acceptance_rate)
```

Shape of observations: (15,)

Initial test - $r(q_0, q_0) = 1.000000$

Running Metropolis algorithm for 100000 iterations...

Proposal distribution: $N(q_{\text{prev}}, D)$ with $D = \text{diag}([0.01, 2e-10])$

Running Metropolis-Hastings MCMC...

Completed 200/100000 iterations, acceptance rate: 0.719
Completed 400/100000 iterations, acceptance rate: 0.759
Completed 600/100000 iterations, acceptance rate: 0.783
Completed 800/100000 iterations, acceptance rate: 0.796
Completed 1000/100000 iterations, acceptance rate: 0.799
Completed 1200/100000 iterations, acceptance rate: 0.808
Completed 1400/100000 iterations, acceptance rate: 0.816
Completed 1600/100000 iterations, acceptance rate: 0.821
Completed 1800/100000 iterations, acceptance rate: 0.827
Completed 2000/100000 iterations, acceptance rate: 0.822
Completed 2200/100000 iterations, acceptance rate: 0.822
Completed 2400/100000 iterations, acceptance rate: 0.821
Completed 2600/100000 iterations, acceptance rate: 0.823
Completed 2800/100000 iterations, acceptance rate: 0.818
Completed 3000/100000 iterations, acceptance rate: 0.818
Completed 3200/100000 iterations, acceptance rate: 0.819
Completed 3400/100000 iterations, acceptance rate: 0.819
Completed 3600/100000 iterations, acceptance rate: 0.818
Completed 3800/100000 iterations, acceptance rate: 0.815
Completed 4000/100000 iterations, acceptance rate: 0.813
Completed 4200/100000 iterations, acceptance rate: 0.814
Completed 4400/100000 iterations, acceptance rate: 0.813
Completed 4600/100000 iterations, acceptance rate: 0.813
Completed 4800/100000 iterations, acceptance rate: 0.813
Completed 5000/100000 iterations, acceptance rate: 0.813
Completed 5200/100000 iterations, acceptance rate: 0.814
Completed 5400/100000 iterations, acceptance rate: 0.813
Completed 5600/100000 iterations, acceptance rate: 0.813
Completed 5800/100000 iterations, acceptance rate: 0.813
Completed 6000/100000 iterations, acceptance rate: 0.813
Completed 6200/100000 iterations, acceptance rate: 0.811
Completed 6400/100000 iterations, acceptance rate: 0.811
Completed 6600/100000 iterations, acceptance rate: 0.812
Completed 6800/100000 iterations, acceptance rate: 0.812
Completed 7000/100000 iterations, acceptance rate: 0.812
Completed 7200/100000 iterations, acceptance rate: 0.811
Completed 7400/100000 iterations, acceptance rate: 0.809
Completed 7600/100000 iterations, acceptance rate: 0.809
Completed 7800/100000 iterations, acceptance rate: 0.809
Completed 8000/100000 iterations, acceptance rate: 0.807

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Completed 8200/100000 iterations, acceptance rate: 0.807
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```

Final acceptance rate: 0.804

Computing posterior values for MAP estimate...

=====

PROBLEM 1.1 RESULTS

=====

MAP Estimate (q_{MAP}):

$$\Phi = -17.375551 \text{ W/cm}^2$$

$$h = 0.001793 \text{ W/cm}^2/\text{°C}$$

$$\pi(u|q_{\text{MAP}})\pi\theta(q_{\text{MAP}}) = 5.74\text{e-}03$$

Posterior Distribution $\pi(q|u)$ Summary:

=====

Φ :

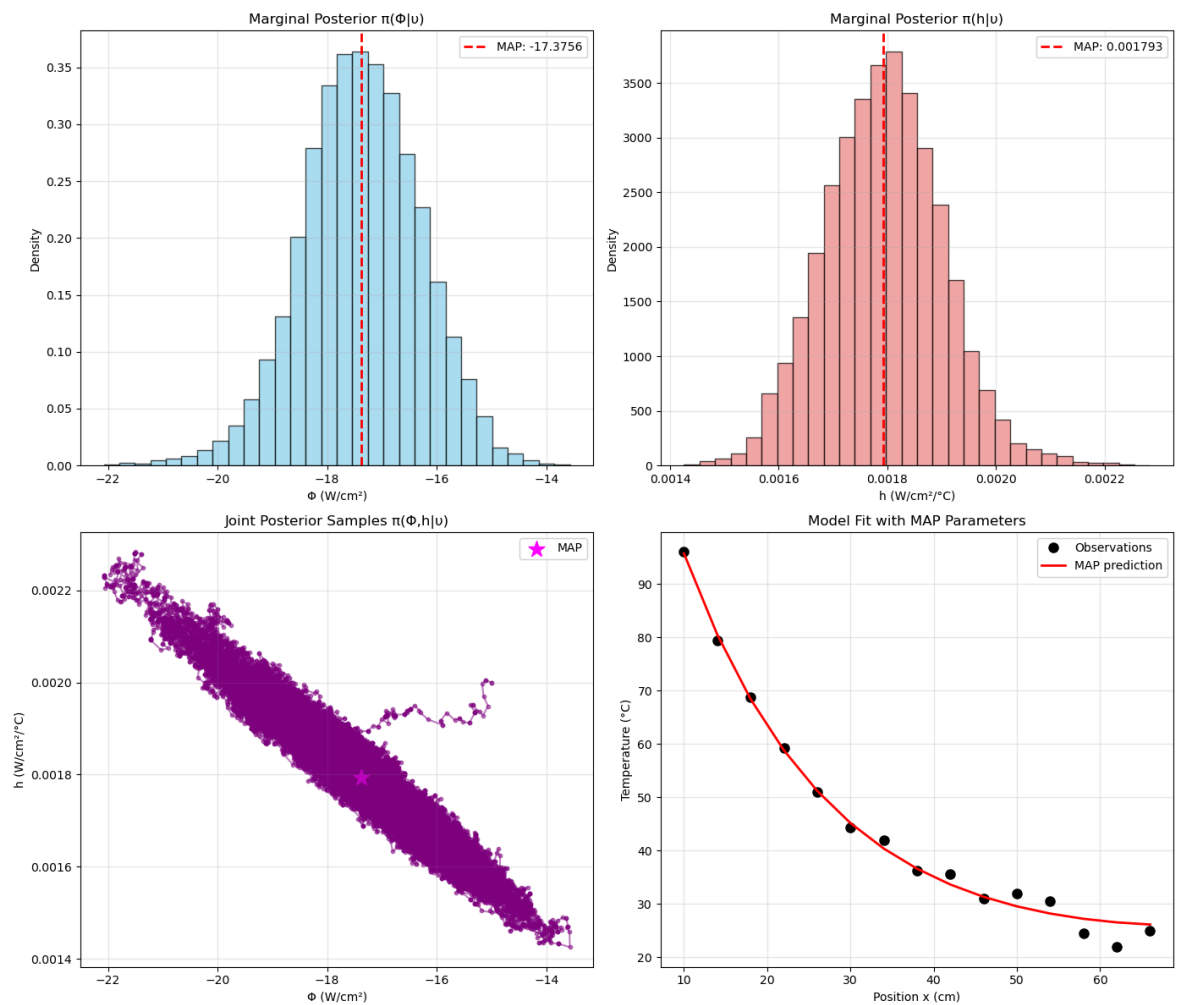
$$\text{Mean} = -17.380237 \text{ W/cm}^2$$

$$\text{Std} = 1.084073 \text{ W/cm}^2$$

$$95\% \text{ CI} = [-19.571328, -15.320156]$$

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h:

Mean = $1.793447\text{e-}03 \text{ W/cm}^2/\text{°C}$ Std = $1.083245\text{e-}04 \text{ W/cm}^2/\text{°C}$ 95% CI = [$1.587000\text{e-}03$, $2.009767\text{e-}03$]

In [11]:

```

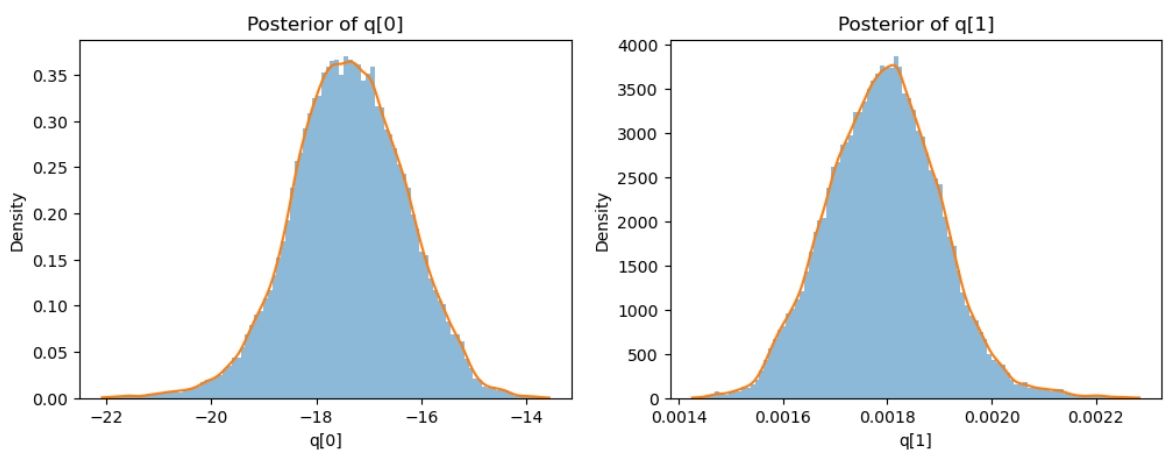
from scipy.stats import gaussian_kde
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
for i in range(2):
    s = samples[:, i]

    # Histogram
    axes[i].hist(s, bins=100, density=True, alpha=0.5)

    # KDE smooth curve
    kde = gaussian_kde(s)
    x = np.linspace(min(s), max(s), 300)
    axes[i].plot(x, kde(x))

    axes[i].set_title(f"Posterior of q[{i}]")
    axes[i].set_xlabel(f"q[{i}]")
    axes[i].set_ylabel("Density")

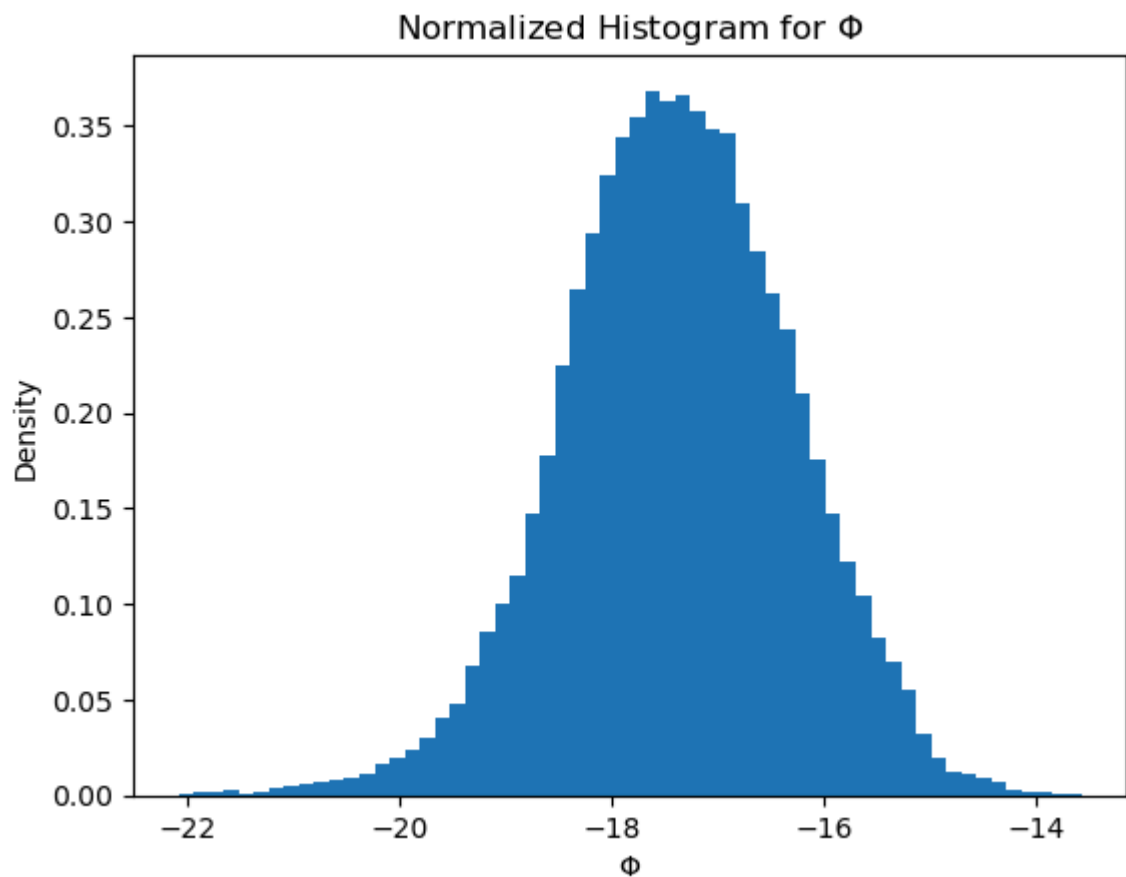
```



In [19]:

```
# Normalized histograms for phi, h
plt.hist(samples[:,0], density = True, bins = 60)
plt.xlabel(r"$\Phi$")
plt.ylabel(r"Density")
plt.title(r"Normalized Histogram for $\Phi$")
```

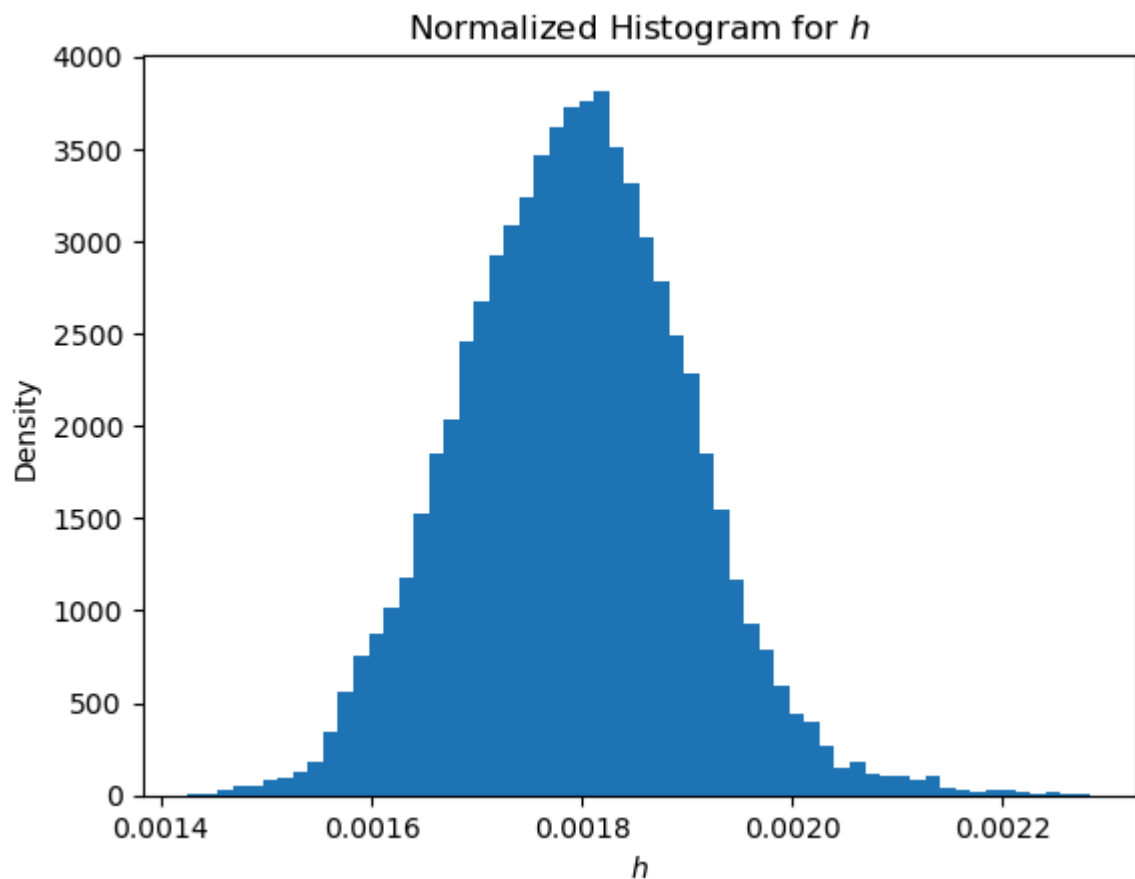
```
Text(0.5, 1.0, 'Normalized Histogram for $\Phi$')
```



In [20]:

```
plt.hist(samples[:,1], density = True, bins = 60)
plt.xlabel(r"$h$")
plt.ylabel(r"Density")
plt.title(r"Normalized Histogram for $h$")
```

```
Text(0.5, 1.0, 'Normalized Histogram for $h$')
```



The new ${}^{22}q_{MAP} = [\Phi = -17.375W / cm^2, h = 0.00179W / cm^2 / ^\circ C]$